

Predicting Exoplanet Habitability from NASA using Machine Learning and Data Analysis

DS-450: Data Science Senior Capstone

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Introduction

The dataset used for this analysis is from the NASA Exoplanet Archive [1] which is an exoplanet catalog for all exoplanets detected by ground- and space-based telescopes. This dataset was used since it contained important key features about a particular exoplanet and its host star which are crucial for conducting an exploratory data analysis on exoplanet data. These features can be used to categorize planets, determine their similarity to Earth, and provide insight into their habitability potential.

This exploratory data analysis will first describe the raw dataset, then highlight the initial data preprocessing steps taken to clean and filter the data. Then secondary data preprocessing will be conducted on the cleaned dataset to handle missing data. This is followed by an analysis into key features of the dataset used to categorize exoplanets, view distributions of key features, and emphasize data visualization.

Upon cleaning the dataset, the target variable which refers to the Earth Similarity Index (ESI), will be extracted from the Habitable Worlds Catalog (HWC) from the Planetary Habitability Laboratory (PHL) sourced from the University of Puerto Rico at Arecibo [3]. The ESI is a multiparameter measure of Earth-likeness for solar or extrasolar planets as a number between zero (no similarity) and one (identical to Earth) [2]. This target variable will be used for the supervised learning models, Random Forest and Gradient Boosted Decision Tree. In the NASA Exoplanet Archive dataset, each observation corresponds to a planet and its associated host star. These identifiers will be used to accurately match and integrate the Earth Similarity Index (ESI), extracted from the Habitable Worlds Catalog, into the cleaned dataset.

Dataset Description

The raw dataset contained 39,235 rows and 92 columns. Key features in the raw dataset were all measured properties observed by ground and space-based telescopes. Initial preprocessing steps were taken to delete redundant features and rename columns. This left 21 columns remaining.

The dataset contained duplicate rows for a single exoplanet where for a particular exoplanet, each duplicate row corresponded to observations conducted by different authors. These duplicate rows contained varying data. To address this issue without arbitrarily removing observations, an outlier detection algorithm was developed to identify

potentially inconsistent measurements within each exoplanet group per column. Exoplanets were grouped by name, and the Z-Score method was then applied to each planet group. The population mean and population standard deviation were computed after excluding missing values. If the standard deviation was less than or equal to 0.01, the variability was considered negligible, and no outliers were flagged for that column. If sufficient variability present, the z-score for each entry was computed using the following formula:

$$z = \frac{x - \mu}{\sigma}$$

where σ is the population standard deviation, μ is the population mean, and x represents the observed entry value.

For most numerical features, a threshold of 3 standard deviations was used in accordance with a 99.7% confidence interval. Observations with absolute Z-scores greater than 3 were flagged as outliers. However, for sensitive stellar parameters such as stellar mass and stellar radius, a stricter threshold of 1 standard deviation was applied to better capture subtle inconsistencies in reported measurements. Any entry flagged as an outlier that also had a “default flag” value of 1 was retained in the dataset, as these measurements are heavily verified by NASA. This allowed for the consolidation of all published observations of exoplanets while preserving the accuracy of default flag entries. Entry values flagged as outliers were automatically replaced with null values. This approach preserves other valid measurements within the same observation while preventing extreme values from skewing statistical analyses. Replacing outliers with null values avoids incorrectly implying a true measurement of zero, which would misrepresent the physical meaning of the data parameter.

After handling outliers, rows corresponding to unconfirmed exoplanets were filtered out if they were classified as a TESS Candidate based on their “solution type.” Additionally, controversial exoplanets with a “planet controversy flag” value of 1 were excluded from the dataset.

After removing the outliers and restricting the data to confirmed exoplanets only, a duplicate row reduction algorithm was created using a custom aggregation function to aggregate numerical columns by the mean and aggregate categorical columns by the mode. This combined the duplicate rows into a single row for a given exoplanet group. This resulted in 6,071 confirmed exoplanets remaining as of January 2026.

The following **Table 1** summary the features used to conduct data preprocessing for producing the cleaned data. This includes their data types, data designation, and percentage of missing data.

Variable Name	Definition	Data Type	Missing Data (%)
default_flag	Describes whether the specific parameters of an exoplanet for a given publication is the most scientifically reviewed by NASA (binary 1 or 0)	float64, nominal	0%
solution_type	Describes whether the exoplanet is confirmed or a TESS candidate	Object, nominal	0%
planet_controversy_flag	Describes whether the exoplanet is controversial (whether its current classification is disputed) or not (binary 1 or 0)	float64, nominal	0%

The following **Table 2** summarizes the key features of the cleaned dataset used for analysis. This includes their data types, data designation, and percentage of missing data.

Variable Name	Definition	Data Type	Missing Data (%)
planet_name	Name of exoplanet	Object, nominal	0%
host_star_name	Name of the host star	Object, nominal	0%
star_number_per_system	Number of stars in the exoplanet's solar system	float64, ratio	0%
discovery_method	The discovery method used to detect the exoplanet	Object, nominal	0%
year_of_discovery	The year the exoplanet was discovered	float64, interval	0%
discovery_facility	The facility (observatory/laboratory/Institution) which detected the exoplanets	Object, nominal	0%
orbital_period_day	The time it takes an exoplanet to complete one revolution around its host star (measured in Earth days)	float64, ratio	5.24%
orbit_semi_major_axis_au	The distance between an exoplanet and its host star (measured in AU – Astronomical Unit which refers to the distance between the Earth and the Sun)	float64, ratio	5.09%
planet_rad_earth_rad	Radius (size) of exoplanet (measured in Earth Radius)	float64, ratio	24.91%
planet_mass_earth_mass	Mass of exoplanet (measured in Earth Mass)	float64, ratio	47.62%
orbital_eccentricity	Describes how elongated (elliptical) the orbit of an exoplanet is.	float64, ratio	14.87%
planet_equil_temp_k	Exoplanet equilibrium (average surface) temperature (measured in terms of Kelvins)	float64, ratio	26.90%
spectral_type	Host star classification which describes a stars luminosity class (whether it is Main Sequence, Sub-Giant, Red Giant, Super Giant, White Dwarf, Brown Dwarf, etc.) and it also describes the star type (whether it is a type M, K, G, F, A, B, O)	Object, nominal	68.57%
stellar_effective_temp_k	Host star's effective (surface) temperature (measured in terms of Kelvins)	float64, ratio	7.68%
stellar_rad_solar_rad	Radius (size) of host star (measured in Sun Radius)	float64, ratio	9.98%
stellar_mass_solar_mass	Mass of host star (measured in Sun Mass)	float64, ratio	0.66%

stellar_surf_grav	Surface gravity of the host star which describes the strength of the gravitational force acting on the surface of the host star	float64, ratio	10.69%
distance_parsec	Describes the distance between the exoplanet and Earth (measured in par seconds)	float64, ratio	2.03%

Dataset Summary Statistics

Data Anomalies – Visualizing Outliers:

After running the outlier detection algorithm, 39,235 rows were flagged as outliers. Out of which, 2,549 rows corresponded to a default flag row. So, these rows were overridden to be kept in the dataset. Therefore, a total of 19,555 rows were found to contain at least one entry flagged as an outlier.

Output visuals were then generated in the form of highlighted tables to showcase exoplanets which contained particularly anomalous outliers. The following tables will highlight the flagged outliers for the planet HD 106315 c.

All entries for planet 'HD 106315 c' in column 'orbital_period_day':

	default_flag	planet_name	orbital_period_day
1958	0	HD 106315 c	1221.270977
1959	0	HD 106315 c	21.056520
1960	0	HD 106315 c	21.057600
1961	0	HD 106315 c	21.057880
1962	1	HD 106315 c	21.057040
1963	0	HD 106315 c	21.056030
1964	0	HD 106315 c	21.056614
1965	0	HD 106315 c	21.057504
1966	0	HD 106315 c	21.056520
1967	0	HD 106315 c	21.056830
1968	0	HD 106315 c	21.057310
1969	0	HD 106315 c	21.058000

Table 3 shows the exoplanet HD 106315 c where the Orbital Period column contained at least one entry flagged as an outlier.

All entries for planet 'HD 106315 c' in column 'stellar_mass_solar_mass':

	default_flag	planet_name	stellar_mass_solar_mass
1958	0	HD 106315 c	nan
1959	0	HD 106315 c	1.154000
1960	0	HD 106315 c	1.070000
1961	0	HD 106315 c	1.080000
1962	1	HD 106315 c	1.091000
1963	0	HD 106315 c	1.024000
1964	0	HD 106315 c	1.070000
1965	0	HD 106315 c	1.119172
1966	0	HD 106315 c	nan
1967	0	HD 106315 c	1.330000
1968	0	HD 106315 c	1.079000
1969	0	HD 106315 c	1.027000

Table 4 shows the exoplanet HD 106315 c where the Stellar Mass column contained one entry flagged as an outlier.

All entries for planet 'HD 106315 c' in column 'stellar_rad_solar_rad':

	default_flag	planet_name	stellar_rad_solar_rad
1958	0	HD 106315 c	1.242700
1959	0	HD 106315 c	1.269000
1960	0	HD 106315 c	1.180000
1961	0	HD 106315 c	1.290000
1962	1	HD 106315 c	1.296000
1963	0	HD 106315 c	1.299000
1964	0	HD 106315 c	1.180000
1965	0	HD 106315 c	1.273595
1966	0	HD 106315 c	1.269000
1967	0	HD 106315 c	1.242700
1968	0	HD 106315 c	1.310000
1969	0	HD 106315 c	1.281000

Table 5 shows the exoplanet HD 106315 c where the Stellar Radius column contained two entries flagged as an outlier.

Data Anomalies – Visualizing Missing Values and Calculating Relevant Parameters:

The following table highlights the missing value percentages for the cleaned dataset. Spectral Type was found to have the largest percentage of missing data with 68.57%.

spectral_type	68.57%
planet_mass_earth_mass	47.62%
planet_equil_temp_k	26.90%
planet_rad_earth_rad	24.91%
orbital_eccentricity	14.87%
stellar_surf_grav	10.69%
stellar_rad_solar_rad	9.98%
stellar_effective_temp_k	7.68%
orbital_period_day	5.24%
orbit_semi_major_axis_au	5.09%
distance_parsec	2.03%
stellar_mass_solar_mass	0.66%
host_star_name	0.00%
planet_name	0.00%
star_number_per_system	0.00%
discovery_method	0.00%
year_of_discovery	0.00%
discovery_facility	0.00%

Table 6 shows the percentage of missing data in order of largest percentage missing to least percentage missing.

0.66% of the stellar mass data was found to be missing. To impute for this missing stellar mass when the orbital semi major axis and orbital period are known, Kepler's Third Law was utilized:

$$a = \sqrt[3]{T^2 M_s} \quad (\text{where } M_s = M_{\text{star}}/M_{\text{sun}})$$

Where a is the orbital semi major axis, T is the orbital period, and M_s is the stellar mass in terms of sun's mass.

0.53% of the stellar mass remained missing, So, a secondary imputation method was used impute for missing stellar mass when the orbital semi major axis and orbital period are unknown, but the stellar surface gravity and the stellar radius are known. This resulted in only 0.41% of stellar mass remaining missing. This method utilized the Universal Law of Gravitation:

$$g = \frac{GM_s}{R_s^2} \quad (\text{where } R_s = R_{\text{star}}/R_{\text{sun}})$$

Where R_s is the stellar radius in terms of sun's radius, M_s is the stellar mass in terms of sun's mass, g is the stellar surface gravity, and G is the gravitational constant $6.67430 \times 10^{-11} \text{m}^3 \text{kg}^{-1} \text{s}^{-2}$.

5.09% of the orbital semi major axis data and 5.24% of the orbital period data were missing, these values were imputed using Kepler's Third Law which reduced the missing value percentages to 0.18%, and 0.44% respectively.

7.68% of stellar effective temperature was missing while 26.90% of the planetary equilibrium temperature was missing. To impute for these missing values when the stellar radius and orbital semi major axis are known. This resulted in the planet equilibrium missing data percentage dropping to 9.55% however the stellar effective temperature remained unchanged. The Zero-Greenhouse Equilibrium Temperature of Exoplanet Equation was utilized:

$$T_{\text{eq}} = T_{\text{eff}}(1 - A)^{1/4} \sqrt{\frac{R_s}{2a}}$$

Where T_{eq} is the planetary equilibrium temperature, T_{eff} is the stellar effective temperature, R_s is the stellar radius in terms of sun's radius, a is the orbital semi major axis, and A is the Bond Albedo constant which is used to describe the amount of light absorbed or reflected in a black-body model which is the model used for determining stellar and planetary temperatures. In this analysis, the Bond Albedo constant is assumed to be 0.5 which means the black body is neither highly dark nor highly reflective. This neutral assumption was chosen to avoid introducing bias or making overly specific assumptions in the absence of detailed albedo measurements for individual exoplanets.

47.62% of planet mass in terms of Earth's mass and 24.91% of planet radius in terms of Earth's radius were missing. Therefore, the Transit Method Mass-Radius Estimation was utilized which reduced the percentage of missing data to 0.12% for both parameters:

$$M_p = R_p^{0.52} \quad (\text{where } M_p = M_{\text{planet}}/M_{\text{earth}} \text{ and } R_p = R_{\text{planet}}/R_{\text{earth}})$$

Where M_p is the planet mass in terms of Earth's mass and R_p is the planet radius in terms of Earth's radius.

A stellar luminosity class was created to be used for further analysis on stellar features. This utilizes Stefan-Boltzmann Equation:

$$L_s = R_s^2 T_{\text{eff}}^4 \text{ where } L_s = L_{\text{star}}/L_{\text{sun}}$$

Where L_s is the stellar luminosity in terms of the Sun's luminosity, R_s is the stellar radius in terms of sun's radius, and T_{eff} is the stellar effective temperature in Kelvin.

A planet radial velocity column was created using the Planet Radial Velocity Formula:

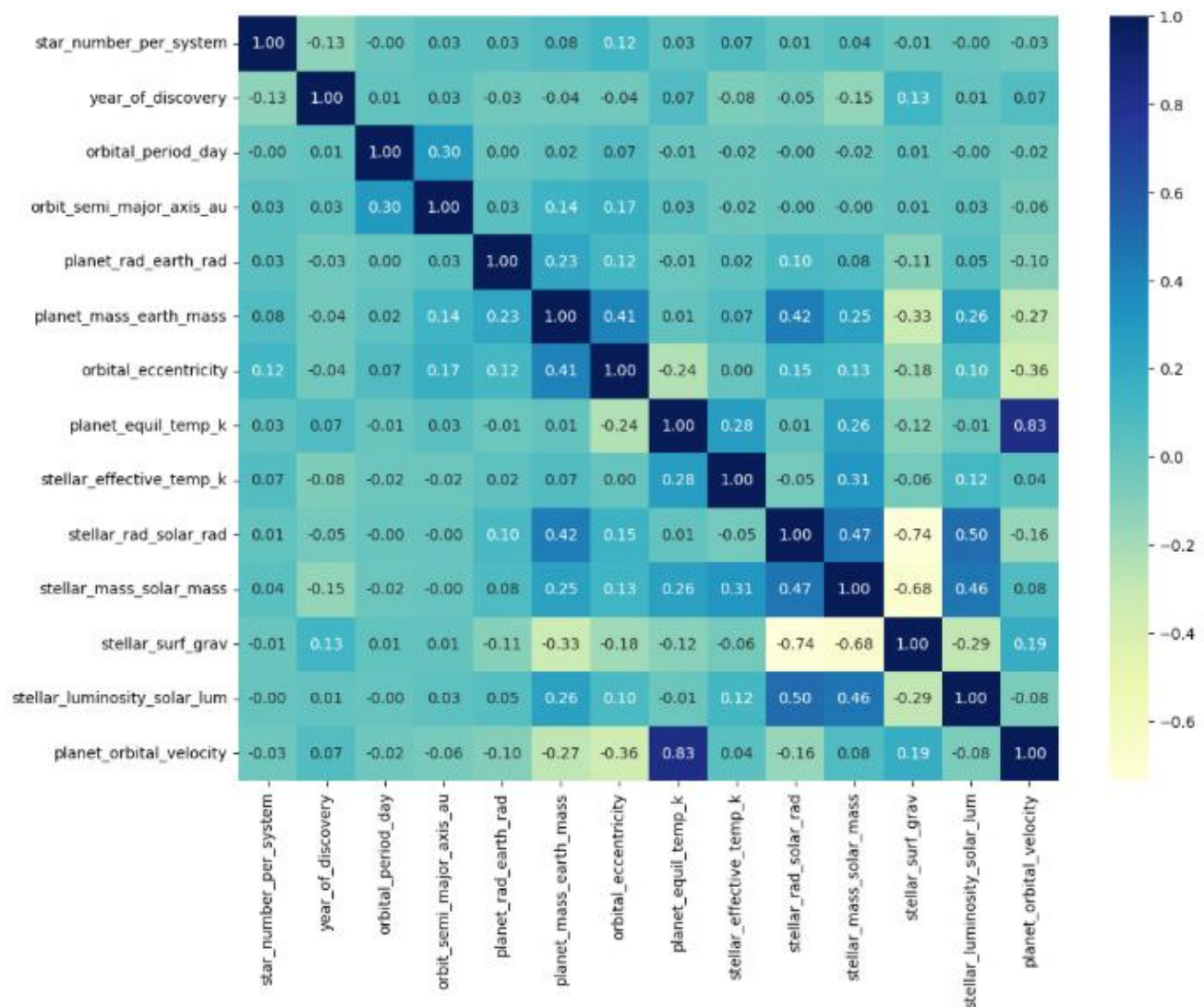
$$v = \frac{GM_p}{a}$$

Where M_p is the planet mass in terms of Earth's mass, a is the orbital semi major axis, G is the gravitational constant $6.67430 \times 10^{-11} \text{m}^3 \text{kg}^{-1} \text{s}^{-2}$, and v is the planet orbital velocity.

Any remaining rows which contained missing data were removed since they consisted of mostly null values for key features which would hinder analysis. The Distance parsec column was also removed due to lack of information available. With the exception of missing rows within the spectral type, orbital eccentricity, stellar luminosity, stellar radius, planet equilibrium temperature, stellar surface gravity, and stellar effective temperature. Further data cleaning will be performed in the future to further address these missing values.

There are currently 6,038 confirmed exoplanets with 20 columns, out of which, 15 are key planetary and stellar features.

Numerical Correlations:



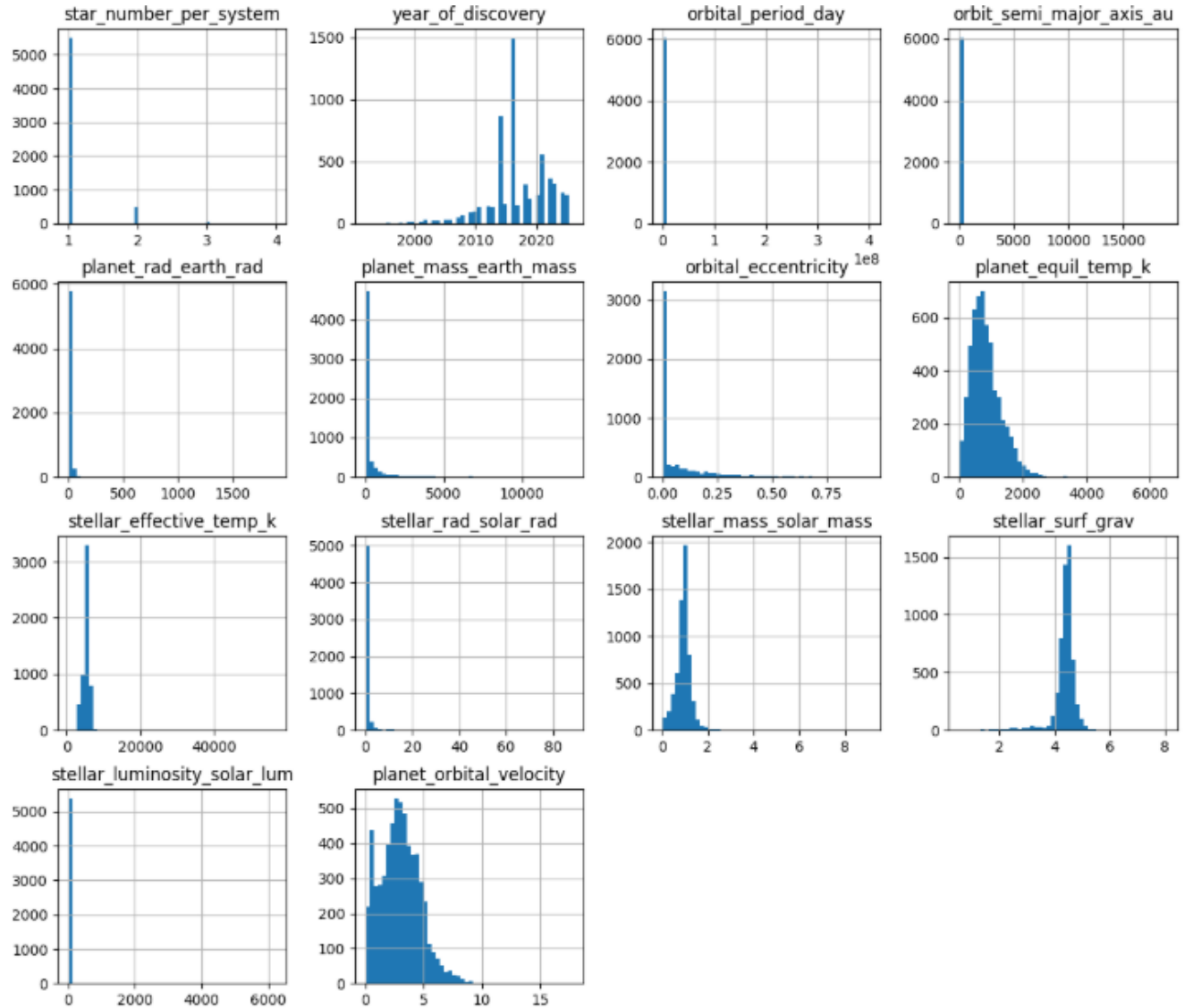
Based on the correlation heatmap for all numerical columns of the dataset, there appears to be minimal to no correlation among a majority of the features. Planet orbital velocity and planet equilibrium temperature however displayed a strong positive correlation of 0.83. Stellar surface gravity displays moderately strong negative correlation with stellar radius and stellar mass with correlation values of -0.74 and -0.68 respectively.

Frequency Distributions and Proportions:

	star_number_per_system	year_of_discovery	orbital_period_day	orbit_semi_major_axis_au	planet_rad_earth_rad	planet_mass_earth_mass	orbital_eccentricity
count	6038	6038	6.04E+03	6038	6038	6038	5164
mean	1.102683	2016.871149	7.09E+04	11.923	9.712176	380.552037	0.076202
std	0.34157	4.886381	5.18E+06	325.260941	43.418697	1117.532336	0.146397
min	1	1992	9.07E-02	0.003857	0.144327	0.0175	0
25%	1	2014	4.12E+00	0.051895	1.874218	1.483045	0
50%	1	2016	1.04E+01	0.09967	2.833239	3.454195	0
75%	1	2021	3.81E+01	0.279476	11.655954	190.458911	0.09
max	4	2026	4.02E+08	19000	1899.339625	13256.62286	0.948
	stellar_effective_temp_k	stellar_rad_solar_rad	stellar_mass_solar_mass	stellar_surf_grav	stellar_luminosity_solar_lum	planet_orbital_velocity	planet_equil_temp_k
count	5594	5460	6038	5571	5.44E+03	6038	5474
mean	5400.201988	1.497106	0.934313	4.375051	6.86E+00	3.014856	833.127985
std	1688.602359	3.824037	0.38142	0.43884	9.94E+01	1.731583	479.672036
min	415	0.0123	0.012617	0.541	8.12E-07	0.004858	5.272973
25%	4923.4	0.774521	0.7757	4.287805	3.58E-01	1.773136	491.4
50%	5561.713636	0.956428	0.947643	4.44114	8.54E-01	2.936925	753.1
75%	5906.875	1.251025	1.082937	4.56803	1.82E+00	4.137818	1087.9
max	57000	88.475	9.1	8.07	6.24E+03	17.840732	6540.30828

Table 7 shows the summary statistics of all numerical columns of the cleaned dataset

The summary statistics referenced in Table 7 reveal that several key planetary and stellar features exhibit right-skewness where the mean exceeds the median. This is particularly evident in parameters such as orbital period, orbital semi-major axis, planetary mass, planetary radius, and luminosity. These distributions are consistent with observational detection bias for short-period, closer distance, massive, larger, and highly irradiated planets. This is expected as more massive and larger planets tend to be easier to detect while brighter stars are easier to measure, and rapidly orbiting planets which are typically closer to their host stars are more abundant.

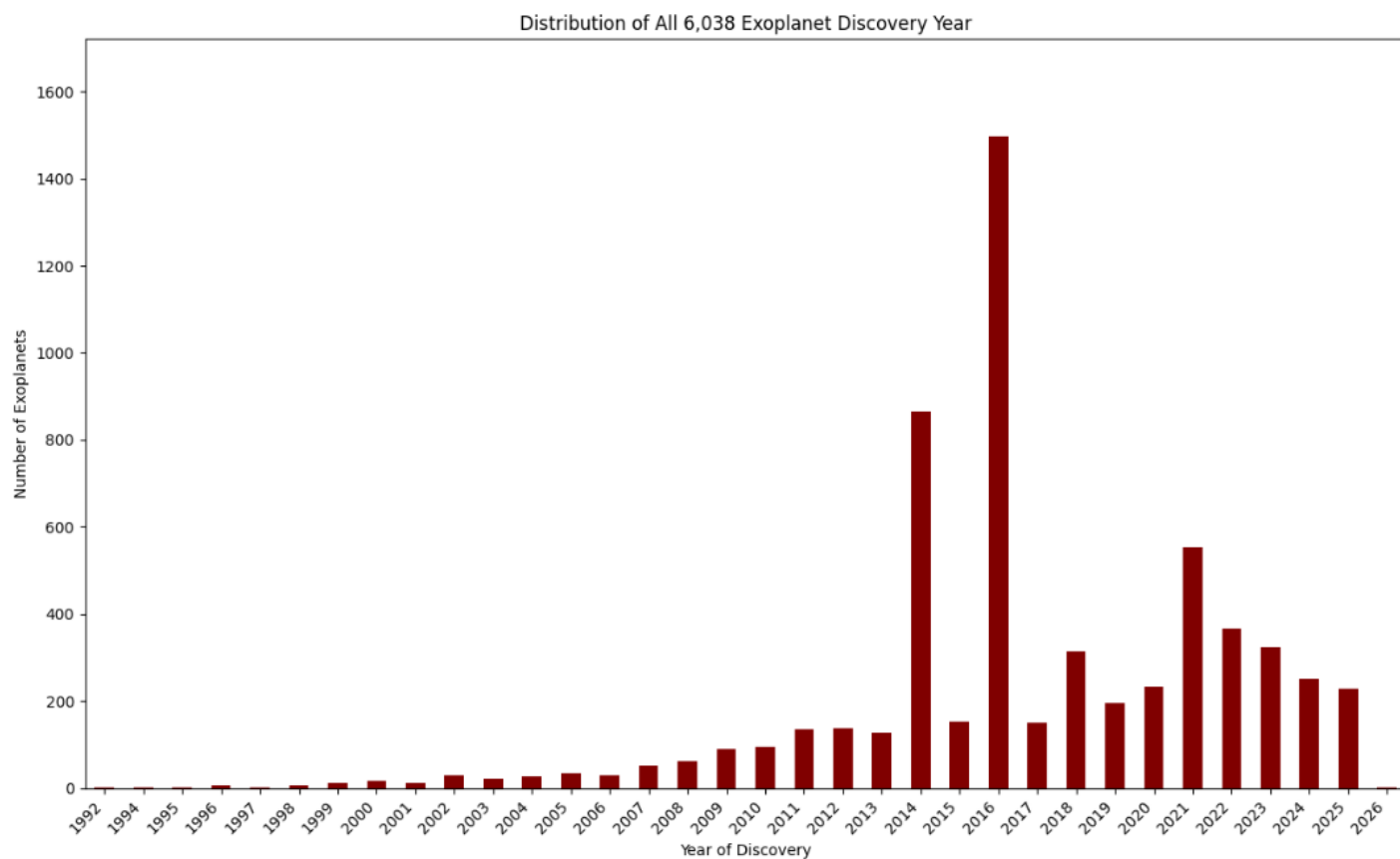


A visualization of histograms for all numerical features of the dataset confirms the findings from the summary statistics. Overall, all parameters exhibit strong right skewness with the exception of discovery year, which is more sporadic with a large spike within the mid-2010s which coincides with the rapid increase in technological advancement of planetary observations. Stellar surface gravity which presents a strong unimodal distribution.

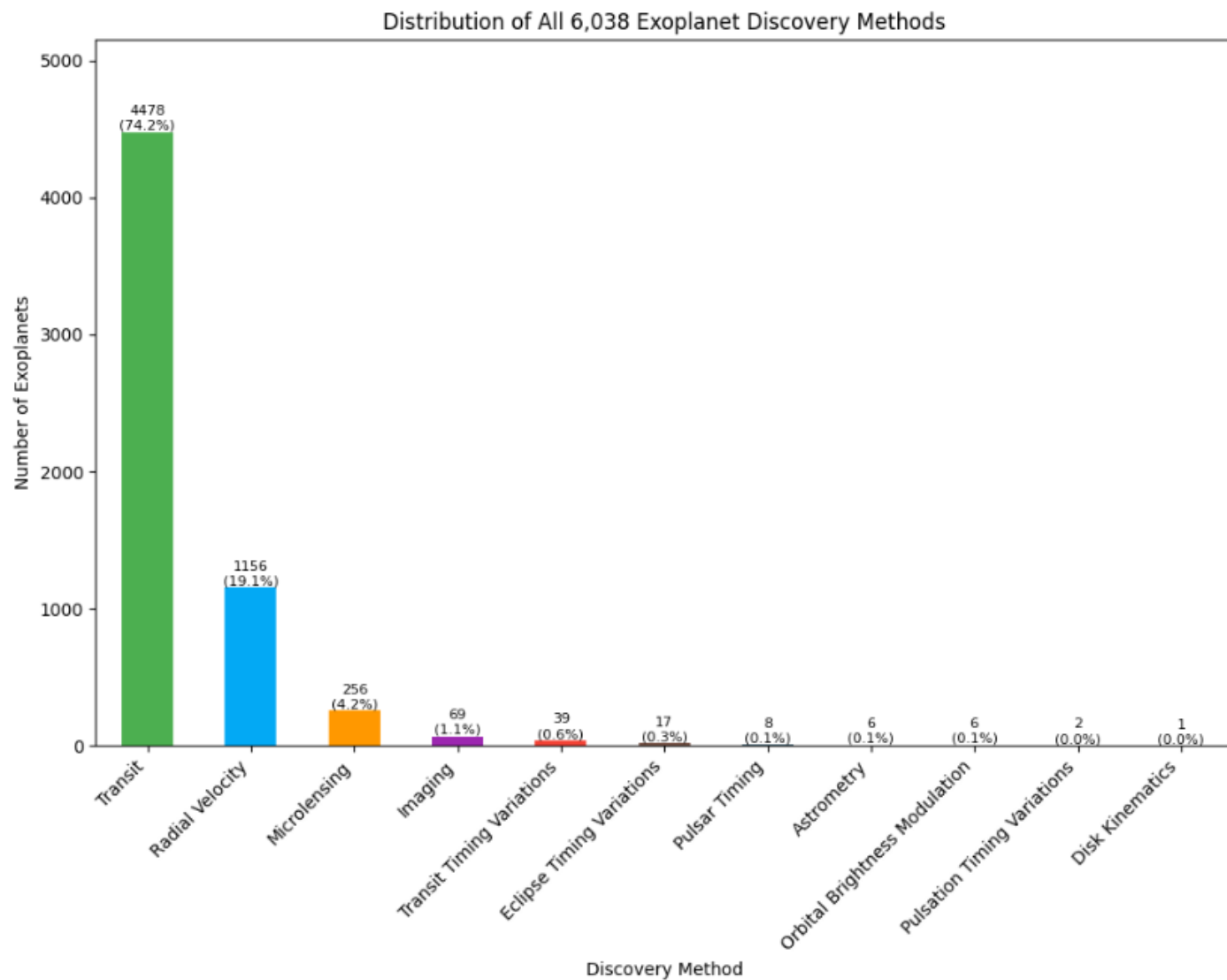
Planet orbital velocity and planetary equilibrium temperature display a moderate right skew but with a wider distribution. Stellar mass exhibits a moderate right skew. However, based on its narrow distribution between 0 and 2, this indicates that a vast majority of host stars exhibit a mass similar to that of the Sun's. The number of stars per system displays a dominant spike at 1 which indicates that a single majority of all extrasolar systems detected

possess a single star like the solar system. Overall, the distributions found within the histograms reflect the expected trends and bias of current astronomical detections.

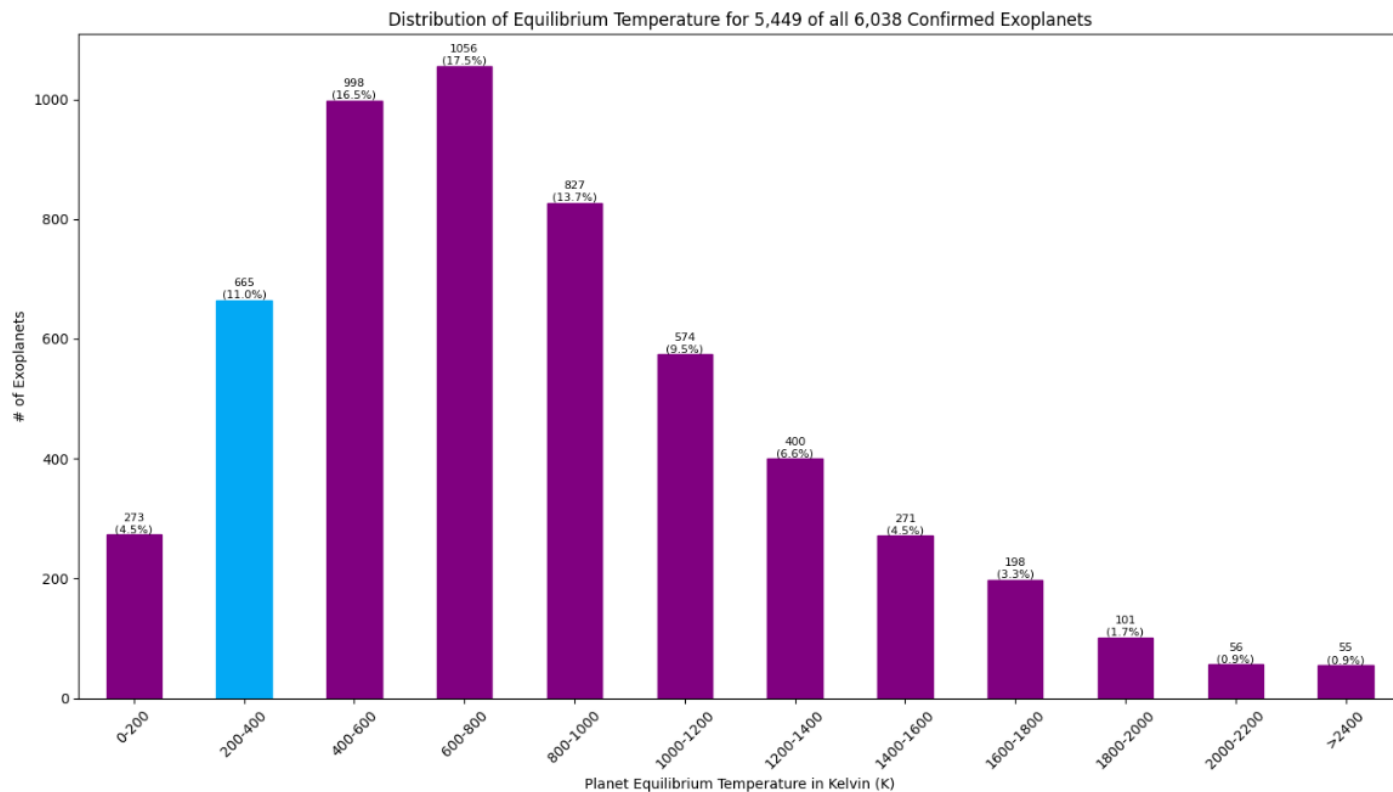
Data Set Graphical Exploration



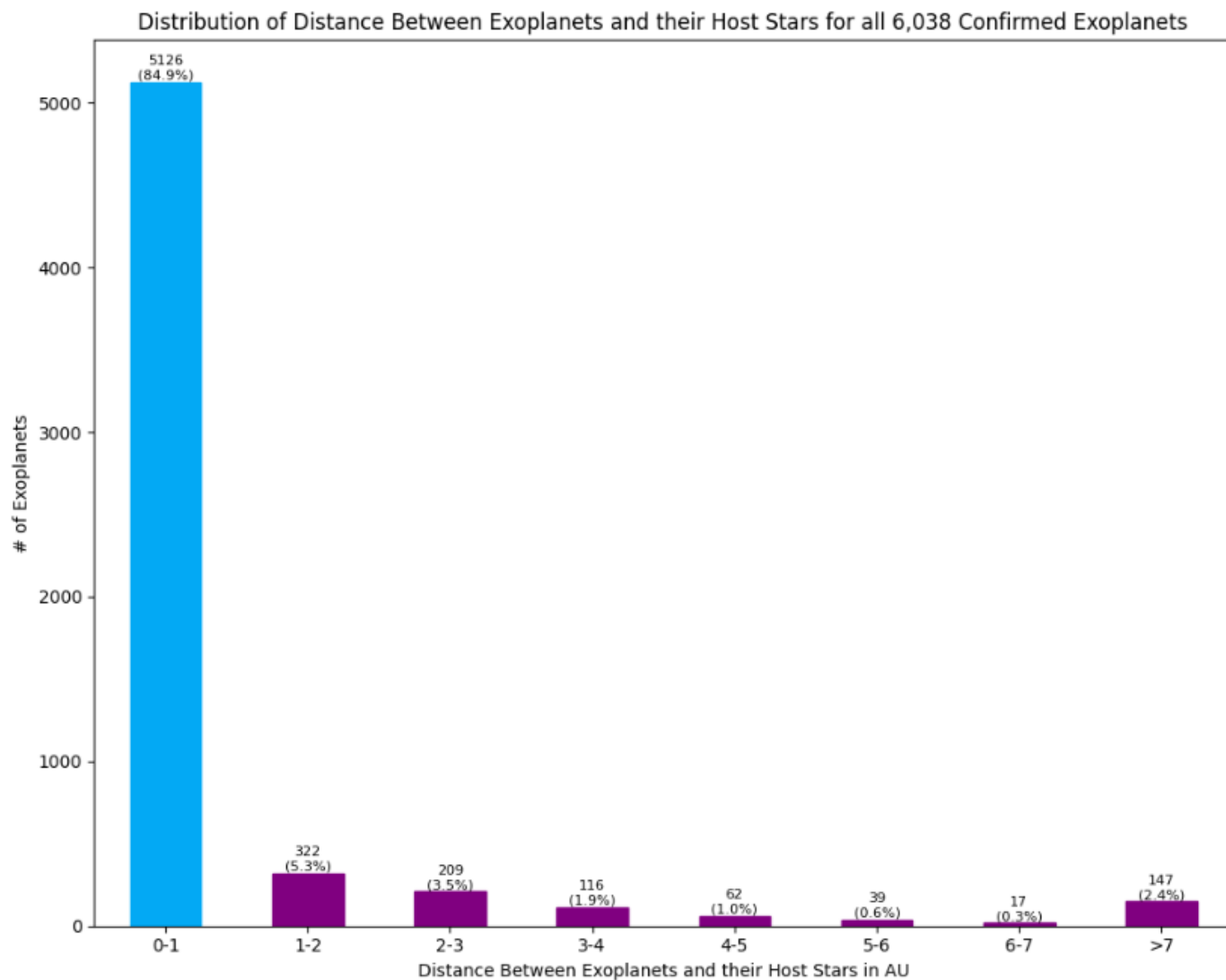
Based on the bar graph to further analyze the distribution of the discovery year, show that a vast majority of exoplanets were discovered within the years 2016 and 2014 and moderately within 2021. Additionally, the overall trend across all years shows a steady increase which was reflected in the prior histogram. This once again reflects the advances in detection methods and space-based missions throughout the years.



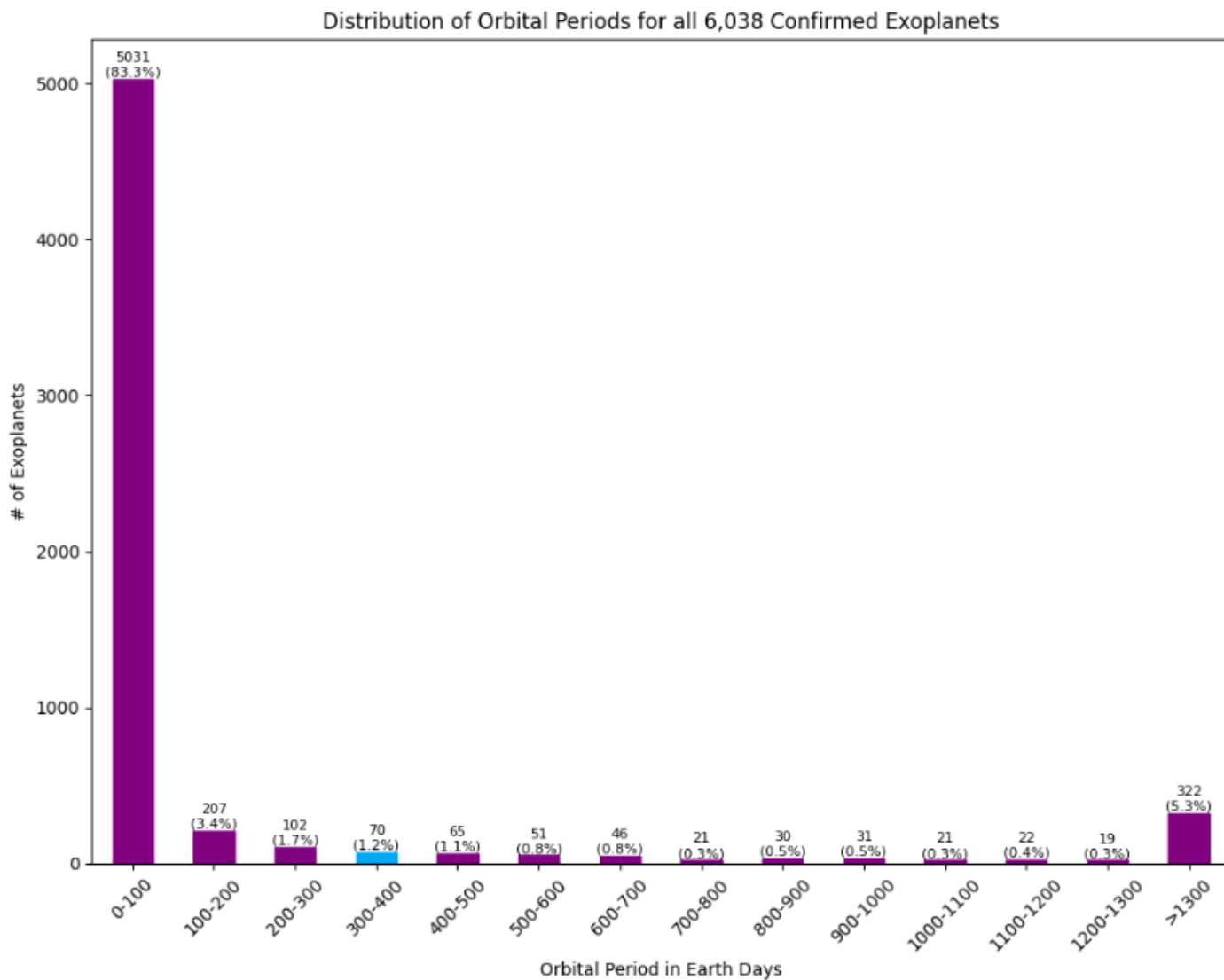
The following bar graph presented above shows that an overwhelming majority of exoplanets were discovered using the transit method at 74,2% followed by radial velocity and microlensing. This highlights the major shift in detection methodologies which prioritizes the Transit method due to its ability to provide insight into the orbital period and planet size.



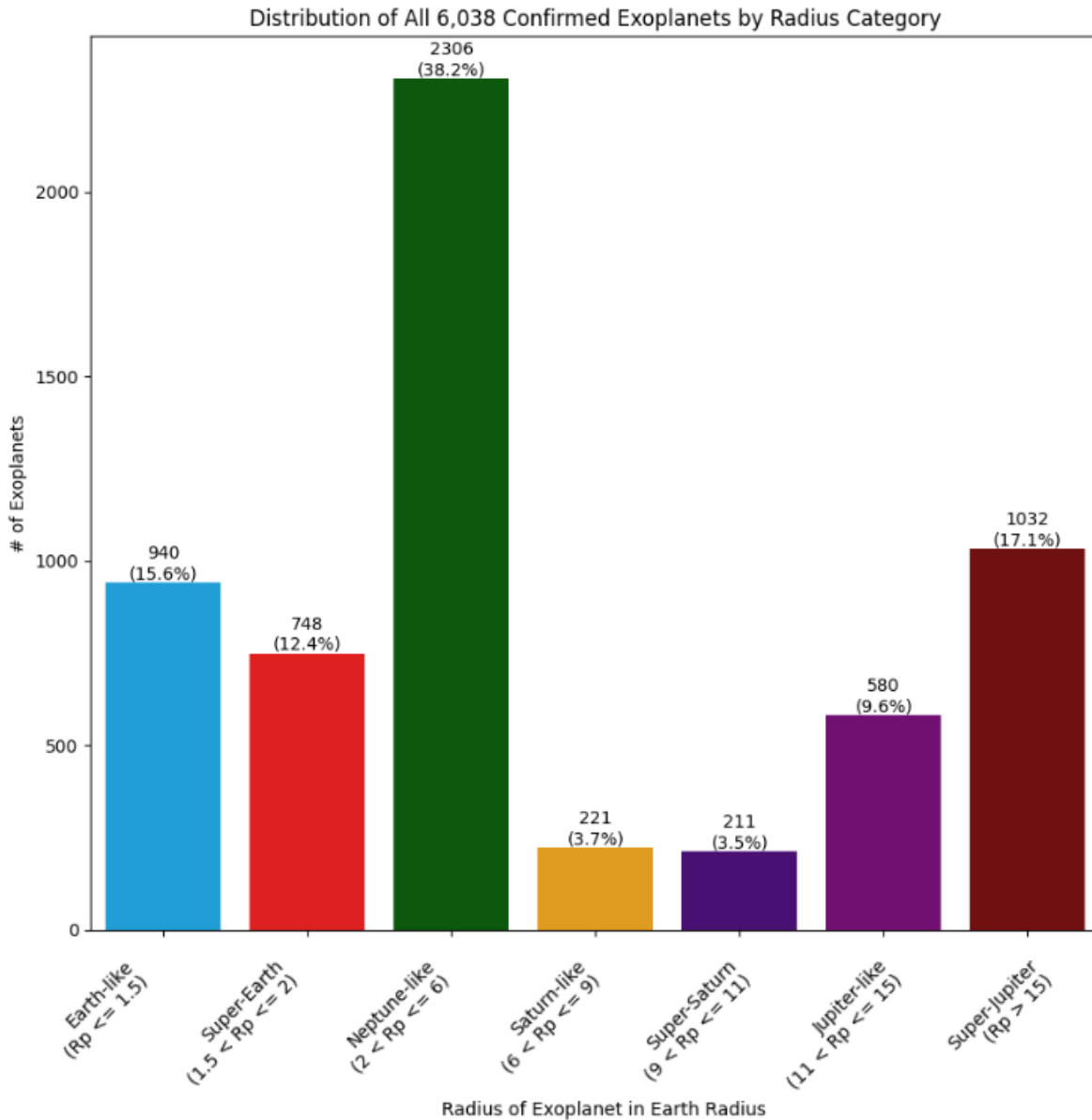
The following bar graph presented above highlights the distribution of planetary equilibrium temperature. Majority of exoplanets tend to be within the 600K-800K range followed by 400K-600K. However, about 11% of all exoplanets have equilibrium (average surface) temperatures within 200K-400K where Earth's equilibrium temperature is approximately 255K. Overall, these insights show that majority of exoplanets tend to be hotter than Earth,



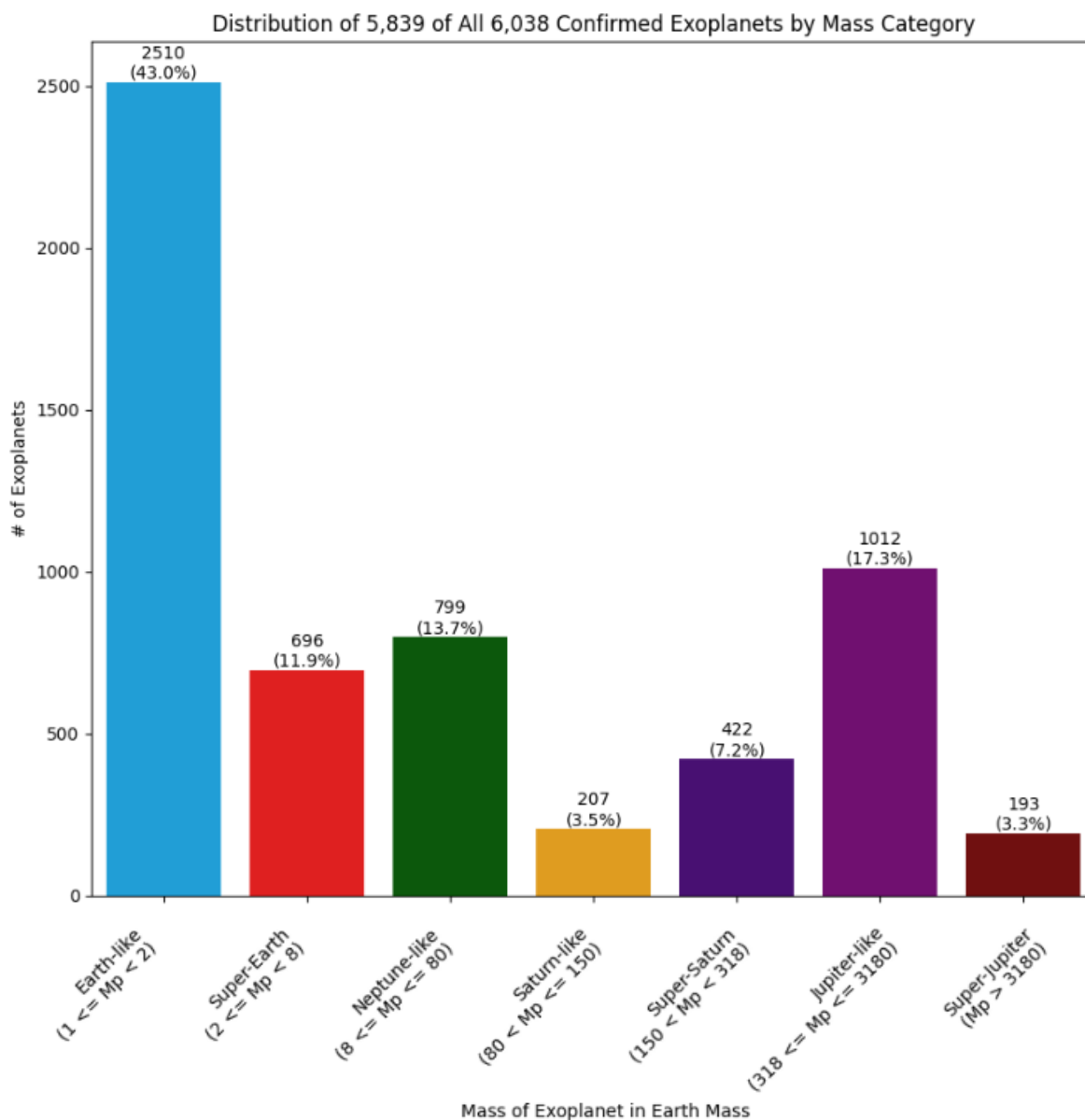
Based on the following bar graph to view the distribution of the distance between an exoplanet and its host star (orbital semi major axis) for all confirmed exoplanets, an overwhelming majority of 84.9% of all confirmed exoplanets were within 1 AU. This provides a promising insight into habitability potential for such exoplanets.



Based on the following bar graph to view the distribution of orbital period for all confirmed exoplanets, a vast majority at 83.3% for all confirmed exoplanets have significantly shorter orbital periods than Earth which is 365.25 days. Only 1.2% of all exoplanets exhibit orbital periods within the same range as Earth. This is reflective of current observational bias due to current detection methodologies used.



Based on the radius characterization of all confirmed exoplanets, a majority were found to be Neptune-Like followed by Super-Jupiter Like, which is expected given that larger exoplanets tend to be easier to detect based on current astronomical observations. However, about 15.6% of all exoplanets were found to be Earth-Like and about 12.4% were found to be Super-Earth Like. This provides positive insights into the number of exoplanets similar to Earth in terms of their size.



Based on the mass characterization for all confirmed exoplanets, a surprising majority at 43% were found to be Earth-Like while 11.9% were found to be Super-Earth Like in mass this provides strong positive insights into identifying potential Earth-Like candidates and their potential for habitability. This is followed by Jupiter Like and Neptune Like.

Summary of Findings

The overall findings from this exploratory data analysis show a strong potential for identifying exoplanets which are Earth Like and mass and radius as well as in their distance between their host stars. Though a majority of exoplanets remain hotter, larger, and shorter in the time they take to orbit their stars compared to Earth and the Sun.

Majority of exoplanets exhibit circular orbits similar to that of Earth which is a positive indicate into Earth similarity. A majority of extrasolar systems consist of one star similar to that of our solar system and a majority of host stars exhibit mass, effective temperature, and luminosity similar to that of the Sun which may indicate a potential abundance for Sun-like stars.

The overall distributions for a majority of numerical features presents strong right skews, this skewness may present major challenges when building supervised learning models since a bias towards a specific value for a given parameter may lead to inaccurate results. To counteract this, a pipeline will need to be utilized for models to help with feature imbalance.

Additionally, remaining missing values for orbital eccentricity, stellar luminosity, stellar radius, planet equilibrium temperature, stelar surface gravity, and stellar effective temperature will need to be addressed through the usage of additional astrophysical formulas. Any rows which still contain missing for these key parameters must be removed. Up to 15% of the rows of the original 6,071 rows may be removed. If a column cannot be imputed, and including it would result in more than 15% of the rows being dropped, then the column will be removed to ensure statistical robustness for model performance for the K-Mean Clustering, Random Forest, and Gradient Boosted Decision Trees models.

The categorization of the spectral type column and imputation of its missing values will need to be addressed by created set ranges based on key stellar factors such as stellar size, effective temperature, and luminosity to determine the star type and luminosity class for a particular host star.

The implementation of the target variable, Earth Similarity Index (ESI) extracted from the Habitable Worlds Catalog, will be conducted by matching the observations of exoplanets and their associated host stars within the Habitable Worlds Catalog and this cleaned dataset.

References

[1] NASA Exoplanet Archive. NASA Exoplanet Archive: Planetary Systems Composite Data. 2026. Available online: <https://exoplanetarchive.ipac.caltech.edu/> (accessed on 11 January 2026).

[2] D. Schulze-Makuch, A. Mendez, A.G. Fairen, et al., “A two-tiered approach to assessing the habitability of exoplanets. *Astrobiology*,” vol. 11, no. 10, 2011.

[3] The Habitable Worlds Catalog (HWC). Planetary Habitability Laboratory (PHL) at the University of Puerto Rico (UPR) Arecibo Composite Data. 2026. <http://phl.upr.edu/hwc> (accessed on 10 February 2026).